Lecture 4. Least Squares: QR Decomposition

Gram–Schmidt, Householder, Givens, conditioning of least squares, underdetermined problems

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4.1 QR decomposition and least squares

Besides Cholesky on the normal equations, least squares can use the QR decomposition (R's `lm`). Let $X \in \mathbb{R}^{n\times p}$ have full column rank ($n \geq p$).

Definition (QR decomposition)

- **Full QR**: $X = QR$, $Q \in \mathbb{R}^{n\times n}$ orthogonal, $R \in \mathbb{R}^{n\times p}$ upper triangular (lower $(n-p)\times p$ block 0). The first p columns of Q are an orthonormal basis of $\mathcal{R}(X)$, the last $n-p$ of $\mathcal{N}(X^\top) = \mathcal{R}(X)^\perp$.
- **Reduced QR**: $X = Q_1 R_1$ with $Q_1 \in \mathbb{R}^{n\times p}$, $Q_1^\top Q_1 = I_p$ (but $Q_1 Q_1^\top \neq I_n$), and $R_1 \in \mathbb{R}^{p\times p}$ upper triangular with positive diagonal.

As $X^\top X = R^\top Q^\top Q R = R_1^\top R_1$, QR of X gives the Cholesky factor of the Gram matrix for free.

Least squares via QR. The normal equations become $R_1^\top R_1 \beta = R_1^\top Q_1^\top y$, i.e. ($R_1$ invertible)

$$R_1 \beta = Q_1^\top y,$$

one back substitution. (Equivalently $\|y - X\beta\|^2 = \|Q_1^\top y - R_1\beta\|^2 + \|Q_2^\top y\|^2$, so RSS $= \|Q_2^\top y\|^2$.) This beats the normal equations because

$$\kappa(X^\top X) = \kappa(X)^2,$$

(singular values of $X^\top X$ are squares of those of X). Standard errors use $(X^\top X)^{-1} = R_1^{-1} R_1^{-\top}$, again by triangular solves.

4.2 Gram–Schmidt

Gram–Schmidt orthonormalization of the columns $x_1, \ldots, x_p$ gives the reduced QR of any full-column-rank X:

(1) $q_1 = x_1 / \|x_1\|_2$.

(2) For $k = 2, \ldots, p$: $v_k = x_k - \sum_{j<k} \langle q_j, x_k \rangle q_j$ and $q_k = v_k / \|v_k\|_2$.

The q_k are orthonormal with $\text{span}\{x_1, \ldots, x_k\} = \text{span}\{q_1, \ldots, q_k\}$. With $r_{jk} = \langle q_j, x_k \rangle$ ($j < k$), $r_{kk} = \|v_k\|_2$, $r_{jk} = 0$ ($j > k$), step (2) reads $x_k = \sum_{j\leq k} r_{jk} q_j$, i.e. $X = Q_1 R_1$.

Classical vs modified. Classical Gram–Schmidt (CGS) projects the original x_k; for nearly collinear columns the computed q's are far from orthogonal. **Modified** Gram–Schmidt (MGS) projects the *current* partially orthogonalized vector, one direction at a time:

$$v_k^{(1)} = x_k, \qquad v_k^{(j+1)} = v_k^{(j)} - \langle q_j, v_k^{(j)} \rangle q_j, \quad j = 1, \ldots, k-1, \qquad q_k = v_k^{(k)} / \|v_k^{(k)}\|.$$

They agree in exact arithmetic; in floating point MGS loses much less orthogonality, though not none. Both cost $\sum_k 4n(k-1) \approx 2np^2$ flops and give only the reduced QR.

Example 4.1 (Where CGS breaks)

Let $x_1 = (1, \epsilon, 0, 0)^\top$, $x_2 = (1, 0, \epsilon, 0)^\top$, $x_3 = (1, 0, 0, \epsilon)^\top$ with $[1 + \epsilon^2] = 1$. Then $r_{11} = [\sqrt{1+\epsilon^2}] = 1$ and $q_1 = x_1$.

- $k = 2$: $r_{12} = q_1^\top x_2 = 1$, $v_2 = (0, -\epsilon, \epsilon, 0)^\top$, $q_2 = (0, -1, 1, 0)^\top/\sqrt{2}$.
- $k = 3$, CGS: with the *original* x_3, $r_{13} = 1$ and $r_{23} = q_2^\top x_3 = 0$, so $v_3 = (0, -\epsilon, 0, \epsilon)^\top$, $q_3 = (0, -1, 0, 1)^\top/\sqrt{2}$, and $q_2^\top q_3 = \frac{1}{2}$: orthogonality is lost.
- $k = 3$, MGS: $v_3^{(2)} = (0, -\epsilon, 0, \epsilon)^\top$ and $q_2^\top v_3^{(2)} = \epsilon/\sqrt{2} \neq 0$, so $v_3 = (0, -\frac{\epsilon}{2}, -\frac{\epsilon}{2}, \epsilon)^\top$, $q_3 = (0, -1, -1, 2)^\top/\sqrt{6} \perp q_2$.

CGS rounds the tiny true $q_2^\top x_3$ away; MGS recomputes it from a vector where it is not negligible. MGS fails in turn for almost parallel columns, e.g. $x_1 = (0.7, 0.7 + \epsilon_{\text{mach}})^\top$, $x_2 = (1/\sqrt{2}, 1/\sqrt{2})^\top$: v_2 is a difference of nearly equal vectors whose direction is rounding noise (with two columns CGS does the same operations). Householder QR, which subtracts no projections, gives an orthogonal Q to machine precision in both cases.

4.3 Householder reflections

Motivation. Orthogonal reflections preserve lengths, so triangularizing X by them solves least squares without squaring the condition number.

Proposition 4.1 (Householder reflector)

For $x, w \in \mathbb{R}^n$ with $\|x\|_2 = \|w\|_2$ and $x \neq w$, let $u = (x - w)/\|x - w\|_2$ and $H = I - 2uu^\top$. Then H is symmetric and orthogonal and $Hx = w$.

Proof. $H^\top = H$ and $H^2 = I - 4uu^\top + 4u(u^\top u)u^\top = I$. With $x^\top x = w^\top w$, $\|x - w\|^2 = 2x^\top x - 2w^\top x$, so

$$Hx = x - 2\frac{(x - w)^\top x}{\|x - w\|^2}(x - w) = x - \frac{x^\top x - w^\top x}{x^\top x - w^\top x}(x - w) = w.$$

■

H reflects across the hyperplane $u^\perp$. Householder QR takes H_1 with $w = \pm\|x_1\|e_1$ to zero column 1 below the diagonal, then $H_2 = \text{diag}(1, \tilde{H}_2)$ on rows 2:n, and so on:

$$H_p \cdots H_2 H_1 X = \begin{pmatrix} R_1 \\ 0 \end{pmatrix}, \qquad Q = H_1 H_2 \cdots H_p.$$

This is the *full* QR. Take $w = -\text{sign}(x_1)\|x\|e_1$, so the first entry of $x - w$ is $x_1 + \text{sign}(x_1)\|x\|$, a sum of like-signed numbers; the other sign gives $x_1 - \text{sign}(x_1)\|x\|$, catastrophically small when $x \approx \|x\|e_1$, and u would be noise.

Implementation and cost. H_j is never formed: $HX = X - 2u(u^\top X)$ costs $4np$. At stage j, $\tilde{u}_j$ costs $3(n - j + 1)$ and applying $\tilde{H}_j$ to the $(n - j + 1) \times (p - j + 1)$ block $4(n - j + 1)(p - j + 1)$; in total

$$\sum_{j=1}^{p} 4(n - j + 1)(p - j + 1) \approx 2np^2 - \tfrac{2}{3}p^3 \text{ flops.}$$

In place: R_1 overwrites the upper triangle and the Householder vectors (first entry 1) are stored below. $Q^\top y$ or Qv apply the reflections in turn; forming Q costs another $2np^2 - \frac{2}{3}p^3$. Householder QR is backward stable; LAPACK `geqrf` (`geqp3` with column pivoting) and R's `qr` use it.

Least squares without forming Q. Apply each reflection to y when computed ($4(n-j+1)$ at stage j); y is overwritten by $Q^\top y = \binom{Q_1^\top y}{Q_2^\top y}$ with no $n \times n$ matrix stored. Then

- $\hat\beta$ solves $R_1\beta = Q_1^\top y$ (the first p entries), by back substitution;
- $\mathrm{RSS} = \|Q_2^\top y\|^2$ (last $n-p$ entries), $\hat\sigma^2 = \mathrm{RSS}/(n-p)$, or RSS/n for the MLE;
- $\widehat{\mathrm{Cov}}(\hat\beta) = \hat\sigma^2 R_1^{-1} R_1^{-\top}$ by triangular solves with the stored R_1.

Total: $2np^2 - \frac{2}{3}p^3 + O(np)$ flops, $O(np)$ memory.

4.4 Givens rotations

A Givens rotation $G(i,k,\theta)$ is the identity except for entries $c, s, -s, c$ in rows and columns i, k ($c = \cos\theta$, $s = \sin\theta$); it is orthogonal. For $y = G^\top x$,

$$y_i = cx_i - sx_k, \quad y_k = sx_i + cx_k, \quad y_j = x_j \ (j \neq i,k),$$

and $c = x_i/\sqrt{x_i^2 + x_k^2}$, $s = -x_k/\sqrt{x_i^2 + x_k^2}$ give $y_k = 0$. $G^\top$ changes only two rows ($6m$ flops). Givens QR zeros one entry at a time, $G_t^\top \cdots G_1^\top X = \binom{R_1}{0}$, in about $3np^2 - p^3$ flops: more than Householder, but suited to sparse or structured matrices (e.g. adding a row to a QR). Each rotation is stored as one number in the zeroed position.

4.5 Conditioning of least squares

For nonsquare X, $\kappa(X) = \sigma_{\max}(X)/\sigma_{\min}(X) = \sigma_{\max}(X)\sigma_{\max}(X^\dagger)$ ($X^\dagger$ the Moore–Penrose pseudoinverse). The condition number of the least squares *problem* is

$$\kappa(X) + \frac{\kappa(X)^2 \tan\theta}{\eta}, \qquad \cos\theta = \frac{\|X\beta\|}{\|y\|}, \quad \eta = \frac{\|X\|\|\beta\|}{\|X\beta\|}.$$

For nearly collinear X and non-small residuals the $\kappa(X)^2$ term dominates, so stable algorithms matter.

Example 4.2

$X = \begin{pmatrix} 1 & 1 \\ 10^{-3} & 0 \\ 0 & 10^{-3} \end{pmatrix}$ has full rank, but with six significant digits $X^\top X = \begin{pmatrix} 1+10^{-6} & 1 \\ 1 & 1+10^{-6} \end{pmatrix}$ rounds to the singular $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$; QR on X itself loses nothing.

method	flops	stability	notes
Cholesky on $X^\top X$	$np^2 + \frac{1}{3}p^3$	least stable	fastest; $X^\top X$ can be updated with streaming data
Householder QR	$2np^2 - \frac{2}{3}p^3$	stable	default in R, MATLAB, Julia
MGS	$2np^2$	stable-ish	gives Q_1 explicitly
SVD	$4n^2p + 8np^2 + 9p^3$	most stable	also handles rank deficiency

No method dominates: Cholesky is twice as fast for $n \gg p$ and needs only $X^\top X$ (which fits in memory when X may not); QR is more accurate.

4.6 Underdetermined least squares

If rank $X < p$ (e.g. $n < p$), the normal equations have infinitely many solutions and need an extra criterion.

- **Minimum ℓ_2 norm**: the solution of $X^\top X\beta = X^\top y$ with smallest $\|\beta\|_2$ is $\hat\beta = X^\dagger y$, equal to $X^\top(XX^\top)^{-1}y$ for full row rank, and the ridge limit $\lim_{\lambda\downarrow 0}(X^\top X + \lambda I)^{-1}X^\top y$.
- **Minimum ℓ_0 "norm"** $\|\beta\|_0 = \#\{j : \beta_j \neq 0\}$: the sparsest solution, i.e. best subset selection (combinatorial).
- **Minimum ℓ_1 norm**: a convex relaxation (basis pursuit, related to the lasso), a linear program that under suitable conditions recovers the sparsest solution.

4.7 Selected exercises

Motivation. Uniqueness of reduced QR, the minimum-norm solution as ridgeless limit, and leverages from Q_1.

Exercise 4.1 (Uniqueness of the reduced QR decomposition)

Let $X \in \mathbb{R}^{n\times p}$ have full column rank.
(a) Show that there is exactly one reduced QR decomposition $X = Q_1R_1$ with $Q_1^\top Q_1 = I_p$ and R_1 upper triangular with positive diagonal, and that Gram–Schmidt produces it.
(b) Householder QR may return an R_1 with negative diagonal entries. How is it converted to the form in (a)?

Solution. [Supplement]

(a) $X^\top X$ is positive definite ($v^\top X^\top Xv = \|Xv\|^2 > 0$ for $v \neq 0$). If $X = Q_1R_1$ as stated, $X^\top X = R_1^\top R_1$ with $R_1^\top$ lower triangular with positive diagonal, so by Theorem 3.1 $R_1^\top$ is the Cholesky factor, determined by X, and $Q_1 = XR_1^{-1}$ is determined too. Existence: Gram–Schmidt (Section 4.2) gives $X = Q_1R_1$ with $r_{kk} = \|v_k\|_2 > 0$ ($x_k \notin \operatorname{span}\{x_1, \dots, x_{k-1}\}$).

(b) Let $S = \operatorname{diag}(\operatorname{sign} r_{11}, \dots, \operatorname{sign} r_{pp})$, nonsingular with $S^2 = I$. Then $X = (Q_1S)(SR_1)$ with orthonormal columns and diagonal $|r_{kk}| > 0$, the unique decomposition of (a): the methods agree up to signs of columns of Q_1 and rows of R_1. ■

Exercise 4.2 (The minimum-norm solution)

Let $X \in \mathbb{R}^{n\times p}$ with rank $X = n < p$ and $y \in \mathbb{R}^n$, and put $\beta^\dagger = X^\top(XX^\top)^{-1}y$.
(a) Show that $\beta^\dagger$ solves the normal equations and is the unique solution of smallest Euclidean norm.
(b) Show that $(X^\top X + \lambda I_p)^{-1}X^\top y = X^\top(XX^\top + \lambda I_n)^{-1}y$ for $\lambda > 0$, and deduce that ridge regression converges to $\beta^\dagger$ as $\lambda \downarrow 0$.

Solution. [Supplement]

(a) $z^\top XX^\top z = \|X^\top z\|^2 > 0$ for $z \neq 0$, so $XX^\top$ is invertible. $X^\top$ is injective, so $X^\top X\beta = X^\top y \iff X\beta = y$. $X\beta^\dagger = y$, and every solution is $\beta = \beta^\dagger + d$ with $Xd = 0$. With $\beta^\dagger = X^\top z$, $z = (XX^\top)^{-1}y$, $\beta^{\dagger\top}d = z^\top Xd = 0$, so

$$\|\beta\|^2 = \|\beta^\dagger\|^2 + \|d\|^2 \geq \|\beta^\dagger\|^2,$$

with equality iff $d = 0$.

(b) For $\lambda > 0$ both $X^\top X + \lambda I_p$ and $XX^\top + \lambda I_n$ are positive definite. Multiplying $X^\top (XX^\top + \lambda I_n) = (X^\top X + \lambda I_p) X^\top$ by $(X^\top X + \lambda I_p)^{-1}$ on the left and $(XX^\top + \lambda I_n)^{-1}$ on the right,

$$X^\top (XX^\top + \lambda I_n)^{-1} = (X^\top X + \lambda I_p)^{-1} X^\top.$$

Apply to y. As inversion is continuous at $XX^\top$, the ridge solution tends to $\beta^\dagger$. ■

Exercise 4.3 (Leverages from the QR decomposition)

Let $X \in \mathbb{R}^{n \times p}$ have full column rank with reduced QR decomposition $X = Q_1 R_1$, and let $H = X(X^\top X)^{-1} X^\top$ be the hat matrix.
(a) Show that $H = Q_1 Q_1^\top$.
(b) Show that the leverage h_{ii} is the squared norm of the i-th row of Q_1, that $0 \le h_{ii} \le 1$, and that $\sum_i h_{ii} = p$.
(c) After a Householder QR, how can all leverages be computed without forming an $n \times n$ matrix?

Solution. [Supplement]

(a) $X^\top X = R_1^\top R_1$ with R_1 invertible, so $(X^\top X)^{-1} = R_1^{-1} R_1^{-\top}$ and

$$H = Q_1 R_1 R_1^{-1} R_1^{-\top} R_1^\top Q_1^\top = Q_1 Q_1^\top.$$

(b) $h_{ii} = e_i^\top Q_1 Q_1^\top e_i = \|Q_1^\top e_i\|^2 \ge 0$, the squared norm of row i of Q_1. With $Q = [Q_1, Q_2]$ orthogonal, $1 = \|Q^\top e_i\|^2 = \|Q_1^\top e_i\|^2 + \|Q_2^\top e_i\|^2 \ge h_{ii}$. And $\sum_i h_{ii} = \operatorname{tr}(Q_1 Q_1^\top) = \operatorname{tr}(Q_1^\top Q_1) = p$.

(c) $Q_1 = H_1 H_2 \cdots H_p \binom{I_p}{0}$: apply the stored reflections, last first, to $\binom{I_p}{0}$, at most $4np$ flops each, so $O(np^2)$ flops and $O(np)$ memory, plus $2np$ for the row norms; H is never formed. ■